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Managing redemption risk: The use of ETFs by open-ended funds☆

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ABSTRACT

We examine how investment funds use exchange-traded funds (ETFs) to manage their liquidity. We find that during the COVID-19 market turmoil, investment funds were selling more ETF shares than any other investor sector. Also, open-ended funds that faced larger outflows in March 2020 sold a higher share of their ETF holdings, consistent with the conjecture of open-ended funds selling ETF shares to raise cash in response to redemption requests. Moreover, ETFs with higher investment fund ownership experienced larger outflows in the primary market. Our findings point to a channel for liquidity contagion from open-ended funds to the ETF market.

1. Introduction

The market for exchange-traded funds (ETFs) in the euro area has experienced significant growth in recent years. According to the European Central Bank's (ECB's) Investment Funds Balance Sheet Statistics, the aggregate total net asset (TNA) value of ETFs in the euro area has increased from EUR 347 billion in 2014-Q4 to EUR 2.1 trillion by 2024-Q4. Within the euro area, investment funds are the largest group of investors in ETFs (Fig. 1(a)). Open-ended funds may invest in ETF shares for a variety of reasons. In addition to the low-cost diversification opportunities provided by ETFs, open-ended funds may also use ETFs for liquidity management purposes (Grill et al., 2018; Sherrill et al., 2020; Cai et al., 2025). While the price of an ETF is not as stable as the value of cash-like instruments, ETFs tend to have higher expected returns and allow for a closer alignment with a fund's portfolio benchmark. At the same time, trading ETF shares may be more efficient than trading the various underlying securities. This way, ETFs could be useful instruments to accommodate flows in a relatively efficient way without materially distorting the composition of the fund's portfolio.

Due to their large footprint in ETF markets, investment funds' trading activity might adversely impact the liquidity of ETFs. For instance, during the onset of the COVID-19 crisis, investment funds sold a large amount of ETF shares, especially when compared with other euro area investor sectors (Fig. 1(b)). This period coincided with increased net redemptions from ETFs and a significant widening of the spread between the share price of the ETF and its net asset value (European Central Bank, 2020). Such developments usually indicate a deterioration of liquidity in the underlying instruments, which is of particular relevance for ETFs investing in relatively illiquid assets, such as corporate bonds (Dannhauser and Hoseinzade, 2022; Koont et al., 2025).

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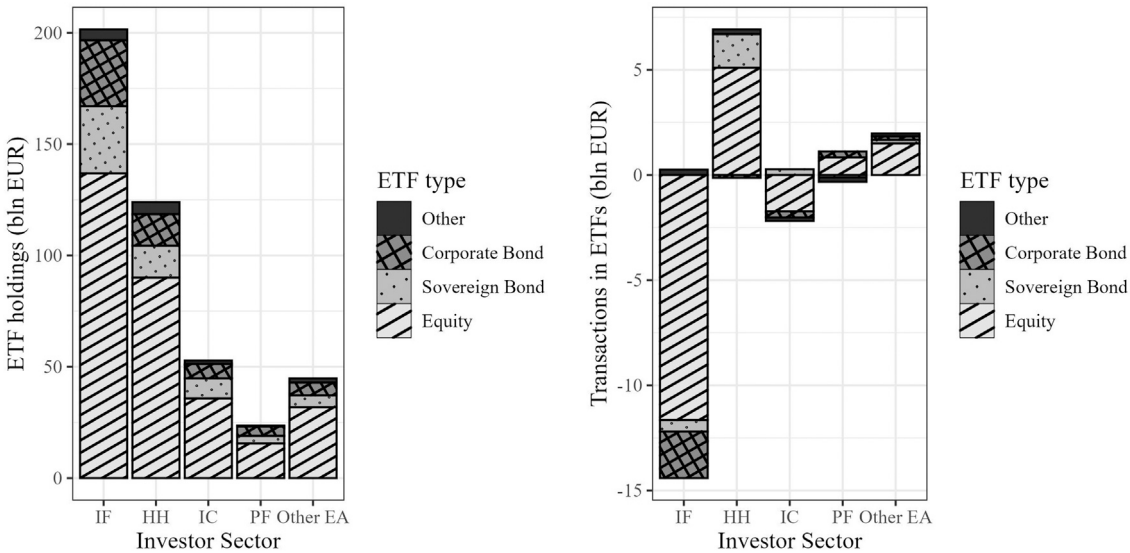


(a) ETF ownership as of 2019-Q4.

(b) Transactions in ETFs during 2020-Q1.

Fig. 1. Ownership and transactions in ETFs during 2020-Q1.
Notes: Panel A shows the value of ETFs held by euro area investors as of December 2019, broken down by investor type and ETF type. The sample of ETFs consists of ETFs belonging to the Undertaking for Collective Investment in Transferable Securities (UCITS) framework. We distinguish between investment funds (IF), households (HH), insurance companies (IC), pension funds (PF), and the remaining investor types are classified as "Other EA". Panel B shows the net transactions by euro area investors during the first quarter of 2020, broken down by investor type and ETF type.
Source: LSEG Lipper, Securities Holdings Statistics.

Given the unique properties of ETFs, we investigate how open-ended funds use ETFs to manage liquidity during periods of stress and how their trading affects ETF markets. We first examine transactions in ETFs during the onset of the COVID-19 crisis by different investor sectors. Using data from the ECB's confidential Securities Holdings Statistics (SHS) database on quarterly ETF holdings by investor sector, we compare changes in ETF holdings across investor sectors during 2020-Q1. Using a similar identification strategy as Allaire et al. (2024) and Fricke et al. (2025), we include ETF fixed effects that allow us to study selling behavior by different investor types within the same ETF within the first quarter of 2020. Second, using more granular data on open-ended funds from LSEG Lipper, we investigate the relation between fund-level redemptions and ETF sales. Specifically, we test whether funds that were subject to larger redemptions also sold a larger fraction of their ETF holdings, which would be indicative of funds passing on their outflows to the ETFs they hold. Third, we study the relationship between ETFs' ownership composition and primary market flows during the COVID-19 crisis using data on daily ETF primary market flows and returns from LSEG Lipper. This allows us to test the hypothesis that ETFs with larger ownership by investment funds were subject to larger net redemptions in the primary ETF market during March 2020.

Our main findings are as follows. First, we find that investment funds were the most run-prone investor type during the COVID-19 crisis. Specifically, we find that investment funds reduced their positions in equity (corporate bond) ETFs by an additional 19 (20) percentage points relative to the household sector during the first quarter of 2020. This is in line with recent findings by Allaire et al. (2024), who show that investment funds were also the largest net sellers of open-ended bond funds during the COVID-19 pandemic. Second, we exploit heterogeneity across open-ended funds and show that those funds that faced larger outflows during March 2020 also scaled down their ETF holdings by a larger amount. Specifically, our tests reveal that a one percentage point increase in outflows during March 2020 implies a 1.6 percentage point additional reduction in ETF holdings. Moreover, we find that funds relied relatively more on ETF shares than on other asset classes when accommodating investor redemptions. This result is consistent with open-ended funds passing on the outflows they faced to the ETF shares they held. To examine the external validity of these results, we extend our sample to the 2019–2024 period. During this extended sample period, we continue finding that (1) open-ended funds facing larger outflows liquidate a larger fraction of their ETF holdings compared to open-ended funds facing smaller outflows; and (2) in response to outflows, open-ended funds liquidate a larger fraction of their ETF holdings relative to their holdings in other asset classes. Third, we find that equity and corporate bond ETFs with higher lagged investment fund ownership faced significantly and substantially larger primary market outflows during the COVID-19 market turmoil, after controlling for lagged ETF performance. Based on a multivariate panel regression model, equity (corporate bond) ETFs with above-median investment fund ownership faced 2.43 (3.75) percentage points additional primary market outflows between March 13 and March 20, as compared to equity (corporate bond) ETFs with below-median investment fund ownership. Overall, these results are consistent with open-ended

funds using ETFs to manage their portfolio liquidity. By selling ETF shares in stressed markets, open-ended funds preserved larger parts of their cash positions and holdings of other assets. At the same time, they were passing on the increased liquidity demand to the ETFs.

Our paper contributes to several strands in the literature. First, we complement recent literature on cross-fund investments by specifically focusing on open-ended funds invested in ETFs (Fricke and Wilke, 2023; Fricke et al., 2025). These papers show that other mutual funds are the most run-prone investors in open-ended fund shares, which is consistent with mutual funds passing on their outflows to the fund shares they hold. As such, these cross-fund holdings give rise to contagion effects within the investment fund sector. Our paper complements this literature by explicitly focusing on the link between open-ended funds and ETFs. The unique characteristics of ETFs could make them more attractive for liquidity management purposes than holdings in other mutual funds that cannot be traded intraday. Consistent with this, our results show that funds holding both ETFs as well as other mutual fund shares scale down their ETF holdings to a larger extent than their holdings in mutual fund shares in response to outflows. Our results therefore highlight an important contagion channel from the open-ended fund sector to ETF markets.

Second, focusing on the use of ETFs for liquidity management, this paper also contributes to the broader literature on liquidity management by open-ended investment funds (Coval and Stafford, 2007; Lou, 2012; Chernenko and Sunderam, 2016; Choi et al., 2020; Jiang et al., 2021; Ma et al., 2022; Dekker et al., 2024). Our analysis is also closely related to the study by Sherrill et al. (2020), which shows that U.S. equity investment funds use ETFs to manage inflows and outflows. We contribute to this literature in three ways. First, we exploit the COVID-19 pandemic to analyze how open-ended funds used ETFs to manage liquidity in response to outflows during a period of significant market stress. Second, we also document the impact of fund ownership on ETF primary market flows during this crisis period. Third, we consider a broad sample of open-ended funds and ETFs covering the entire euro area. The sample includes mixed asset funds, which are the largest group of investment funds holding ETF shares, as well as their investments in bond and equity ETFs. In contrast, Sherrill et al. (2020) only consider equity mutual funds.

Finally, our paper also connects with the literature on the impact of ETF creations and redemptions on underlying asset markets (Dannhauser and Hoseinzade, 2022; Rhodes and Mason, 2023; Shim and Todorov, 2023; Holden and Nam, 2024; Brogaard et al., 2025; Koont et al., 2025). Shim and Todorov (2023) show that authorized participants (APs) can act as a buffer between ETFs and underlying asset markets during times of stress. Namely, when an AP receives the underlying assets in response to ETF share redemptions, the AP faces an incentive to keep these assets in inventory to avoid costly fire sales. Consequently, ETF share redemptions do not translate one-to-one in fire sales of the underlying assets. On the other hand, Koont et al. (2025) argue that inclusion in redemption baskets may in fact reduce the liquidity of the corresponding asset during times of stress. Namely, when an AP receives assets following an ETF share redemption and keeps them in inventory, the AP may become reluctant to buy more of the same assets in secondary markets. This reduction in market-making activity by APs may lead to a deterioration in secondary market liquidity of the corresponding assets. While the impact of ETF primary market flows on underlying asset markets is beyond the scope of our paper, our analyses contribute to this literature by providing new insights in the main drivers of ETF primary market flows.

The remainder of the paper is structured as follows. In Section 2, we describe our dataset and provide summary statistics. In Section 3, we study differential selling behavior by different investor types that invest in the same ETF. In Section 4, we examine the link between the magnitude of outflows faced by open-ended funds during March 2020 and the magnitude of ETF sales by the same open-ended funds. In Section 5, we discuss whether open-ended funds relied more on ETFs relative to other asset holdings to meet redemption requests. In Section 6, we examine the impact of the composition of ETFs' investor base on primary market flows. We conclude in Section 7.

2. Data description

2.1. ETF sample

We obtain daily primary market flows and returns between January 2019 and June 2020 from LSEG Lipper for equity and bond ETFs falling under the Undertakings for Collective Investment in Transferable Securities (UCITS) Directive. The UCITS Directive is a regulatory framework for investment funds in the European Union.[1] We exclude ETFs with TNA below 5 million euros and ETFs that are less than three months old. The combined TNA of the resulting sample of ETFs is equal to 789 billion euros as of December 2019. For comparison, the aggregate TNA by euro area ETFs as of December 2019 amounts to 857 billion euros according to the ECB's Investment Funds Balance Sheet Statistics,[2] implying that LSEG Lipper's coverage of UCITS ETFs is fairly complete.

We merge these data with information on the ownership composition from the ECB Securities Holdings Statistics (SHS) database. We restrict the sample to ETFs for which ownership information from SHS covers at least 75% of the ETF's TNA.[3] We also remove observations where total ownership based on SHS-holdings is more than 5% larger than an ETF's TNA. The difference between an ETF's TNA and the holdings provided by the SHS database is attributed to investors located outside the euro area (which we refer to

[1] Directive 2009/65/EC of the European Parliament and of the Council of July 13, 2009 on the coordination of laws, regulations, and administrative provisions relating to undertakings for collective investment in transferable securities (UCITS).

[2] https://data.ecb.europa.eu/data/datasets/IVF/IVF.Q.U2.N.TE.L30.A.1.Z5.0000.Z01.E

[3] The SHS database covers holdings by investors domiciled in the euro area. In addition, the SHS database partially covers holdings by non-resident investors that make use of euro area custodians. Finally, the SHS database has also started covering some non-euro area EU countries. See Boermans (2025) for more information on the SHS database, as well as its applications in academic research.

Table 1
Summary ETF sample.

Panel A. Asset Type						
		Equity	Bond Corporate	Bond Sovereign	Bond Other	Total
Number of funds		404	57	116	17	594
TNA (EUR bln)		115	37	38	1	190
Panel B. Domicile						
	France	Germany	Ireland	Luxembourg	Other	Total
Number of funds	50	78	136	327	3	594
TNA (EUR bln)	15	21	58	97	0	190
Panel C. Geographical Focus						
	Europe	Global	Emerging Markets	US	Other	Total
Number of funds	225	87	37	95	150	594
TNA (EUR bln)	85	25	14	36	30	190

Notes: This table provides a summary of our sample of ETFs as of December 2019. This sample covers equity and bond ETFs that are regulated under the Undertaking for Collective Investment in Transferable Securities (UCITS) framework. Panel A shows the number of ETFs and aggregate total net assets (TNA) broken down by the asset class to which the ETF allocates. Panel B shows the number of ETFs and aggregate TNA broken down by ETFs' country of domicile. Finally, Panel C shows the number of ETFs and aggregate TNA broken down by ETFs' geographical investment focus.

Table 2
Ownership profile of ETFs as of December 2019.

	Equity ETFs	Corporate Bond ETFs	Sovereign Bond ETFs
Banks	1.67	1.78	1.94
Government	1.77	2.53	2.47
Households	24.35	19.85	22.52
IC	10.08	9.09	11.38
IF	41.51	41.06	39.65
NFC	3.23	2.51	2.3
OFC	2	1.7	1.34
PF	3.7	5.24	4.53
Non-EA	11.67	16.23	13.86

Notes: This table shows the average composition of the investor base of the ETFs in our sample weighted by these ETFs' Total Net Assets (TNA), while distinguishing between ETF types (equity ETFs, corporate bond ETFs, and sovereign bond ETFs). We consider the following investor types: Banks, Government (including social security funds), Households, insurance companies (IC), investment funds (IF), non-financial corporations (NFC), other financial corporations (OFC), pension funds (PF), and non-euro area investors (Non-EA).

as "non-EA" investors). After applying these additional filters, which are in line with Fricke et al. (2025), the aggregate TNA of the resulting sample of ETFs is equal to 190 billion euros as of December 2019. Table 1 shows the composition of the resulting sample of ETFs. Overall, our sample includes 404 equity and 190 bond ETFs accounting for respectively 115 and 76 billion euros in TNA by December 2019 (Panel A). We consider a bond ETF to be a corporate bond ETF if the ETF's portfolio consists for at least 50% of corporate bonds. Likewise, bond ETFs allocating at least 50% to sovereign bonds are considered to be sovereign bond ETFs. We classify the remaining bond ETFs as "Other" bond ETFs. The majority of our ETF sample is domiciled in Ireland and Luxembourg, both in terms of the number of ETFs and TNA (Panel B). Panel C shows that most of the ETFs in our sample have a European investment universe. In addition, a substantial number of ETFs has a global focus, focuses on emerging market assets, or on the United States. Finally, the Other category primarily includes ETFs with a single-country focus other than the U.S. In the remainder of the paper, we restrict attention to equity, corporate bond, and sovereign bond ETFs, thereby excluding bond ETFs that are neither classified as corporate bond ETFs nor as sovereign bond ETFs.

Table 2 shows the ownership composition of the ETFs in our sample. For all three types of ETFs, the investment fund sector is the largest owner type. Other large investor sectors include households, insurance companies, pension funds, and investors located outside the euro area (which we refer to as "non-EA" investors). Hence, the ownership profile of the ETFs in our sample resembles the ownership profile of open-ended funds in the euro area (Allaire et al., 2024). Note that the SHS database generally does not contain information on securities holdings by investors outside the euro area. As such, we are unable to identify the sectoral composition of holders outside the euro area. We therefore ignore changes in holdings by investors located outside the euro area and focus on investment funds, households, insurance companies, and pension funds. We allocate the remaining types of investors located in the euro area (hence excluding non-EA investors) to a residual category labeled "Other EA".

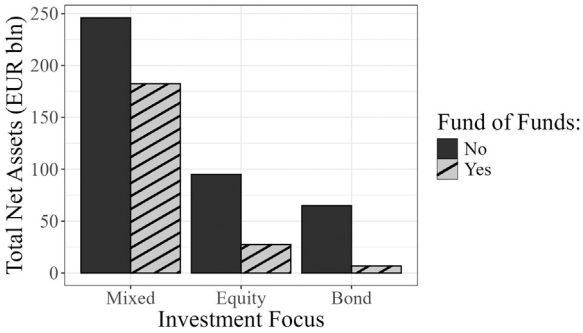


Fig. 2. Summary of open-ended funds investing in ETFs as of 2019-Q4.
Notes: This figure shows the total net assets of open-ended funds holding ETFs, broken down by investment focus (mixed funds, equity funds, or bond funds). We also distinguish between funds of funds and regular mutual funds.
Source: LSEG Lipper.

Table 3
Summary OEF sample.

Panel A. Domicile							
	France	Germany	Ireland	Luxembourg	UK	Other	Total
# funds	154	215	110	778	264	511	2030
TNA (€bln)	27	37	49	266	112	133	622
Panel B. Geographical Focus							
	Europe	Global	Markets	Emerging UK	US	Other	Total
# funds	133	1739	45	31	21	62	2030
TNA (€bln)	34	525	22	16	6	19	622
Panel C. Portfolio composition							
Asset Type			Mean	St. Dev.	5%	50%	95%
ETF			19.83	23.38	0.81	10.47	77.91
Mutual Funds			36.9	33.62	0	29.86	91.24
Equity			19.41	30.23	0	0.02	91.3
Sovereign Bond			7	14.55	0	0	38.41
Corporate Bond			8.54	17.71	0	0	48.2

Notes: This table provides a summary of our sample of open-ended funds. Panel A shows the number of open-ended funds as well as aggregate Total Net Assets (TNA) broken down by funds' country of domicile. Panel B shows the number of open-ended funds as well as aggregate TNA broken down by geographical investment focus. Finally, Panel C summarizes the portfolio compositions of the open-ended funds in our sample. For each panel, we first compute the cross-sectional statistics for each month separately, after which we take the time-series average.

2.2. Open-ended fund sample

We also obtain data for open-ended funds from LSEG Lipper over the period spanning from January 2019 through June 2024. As we are interested in the use of ETFs to manage redemption requests, we only select open-ended UCITS funds holding ETFs. As is standard in the literature, we drop observations for open-ended funds with an age below one year. Fig. 2 shows that the resulting sample predominantly consists of mixed asset funds, followed by equity and bond funds. We discard any remaining open-ended funds with a different investment focus. Also, a substantial fraction of our sample consists of funds of funds.

Table 3 shows that most open-ended funds are domiciled in Luxembourg (Panel A). Moreover, the vast majority of these funds has a global investment focus, both in terms of TNA, as well as the number of funds (Panel B). Panel C shows a summary of the portfolio composition of the open-ended funds in our sample. Funds on average allocate 19.83% to ETFs, but there is large heterogeneity across funds. Apart from ETFs, funds on average allocate large fractions of their portfolios to other mutual fund shares, which intuitively makes sense given that a substantial part of our sample consists of funds of funds.

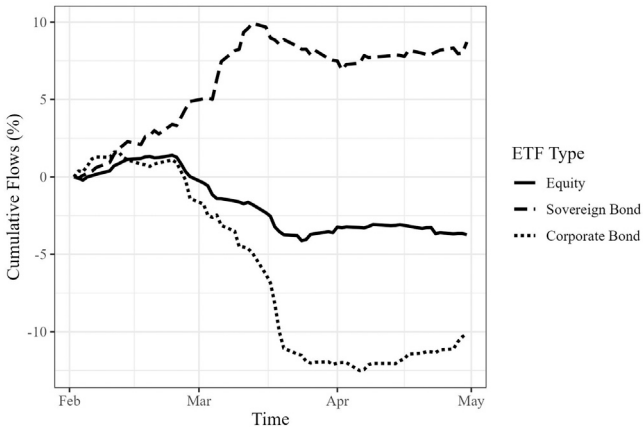


Fig. 3. ETF primary market flows during 2020-Q1.
Notes: This figure shows cumulative primary market flows for all UCITS ETFs, broken down by ETF type. The starting date is February 1, 2020.
Source: LSEG Lipper.

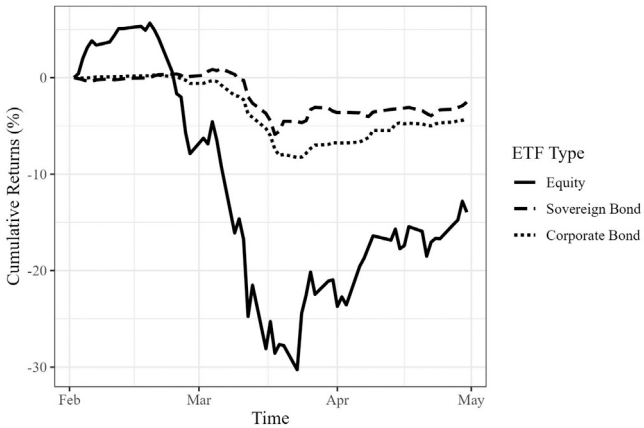


Fig. 4. ETF performance during 2020-Q1.
Notes: This figure shows aggregate cumulative returns for all UCITS ETFs, broken down by ETF type. The starting date is February 1, 2020.
Source: LSEG Lipper.

3. ETF selling by different investor types

Fig. 3 shows ETFs' primary market flows during the COVID-19 pandemic, broken down by ETF type. As shown in Fig. 3, both equity and corporate bond ETFs experienced large net redemptions during this crisis, whereas sovereign bond ETFs faced net inflows in the primary market. Fig. 4 shows the cumulative performance of ETFs as of February 1, 2020, broken down by ETF type. It follows that equity ETFs faced the largest losses, with aggregate cumulative returns ultimately reaching −30%. Corporate bond and sovereign bond ETFs faced substantially milder losses, in line with findings for open-ended funds by Allaire et al. (2024). Still, despite the milder losses for corporate bond ETFs relative to equity ETFs, cumulative primary market outflows during the same period were substantially larger for corporate bond ETFs.

We next study ETF transactions by different investor types in more detail. For each ETF in our sample, we construct the quarterly flow by each investor type using SHSS data:

$$f_{i,q}^{j} = \frac{Transactions_{i,q}^{j}}{MV_{i,q-1}^{j}}. \tag{1}$$

Here, $f_{i,q}^{j}$ denotes the (secondary market) flow of investor type j in ETF i during quarter q. The variable $Transactions_{i,q}^{j}$ denotes the net transactions in ETF i during quarter q by investor type j, recorded at transaction value. Moreover, $MV_{i,q}^{j}$ denotes the market value of the aggregate position of investor type j in ETF i at the end of quarter q. Because the denominator contains the lagged market value that investor type j held in ETF i, $f_{i,q}^{j}$ should be interpreted as the relative quarterly change in the aggregate position of investor type j.

Table 4
ETF trading by different investor sectors.

Dependent variable:	Flow in ETF i by investor type j in 2020-Q1			
ETF type:	All	Equity	Corporate Bond	Sovereign Bond
	(1)	(2)	(3)	(4)
Households	0.106	0.185*	0.198*	−0.189
	(1.69)	(2.56)	(2.02)	(−1.07)
Insurance Companies	0.174*	0.159	0.112	0.275
	(2.30)	(1.80)	(0.850)	(1.32)
Pension Funds	0.281**	0.443**	0.164	−0.075
	(2.64)	(2.67)	(1.39)	(−0.485)
Other EA	0.576**	0.787**	0.380*	−0.034
	(5.67)	(6.17)	(2.02)	(−0.157)
ETF FE	Yes	Yes	Yes	Yes
Observations	2,337	1,598	247	492
# ETFs	572	399	57	116
Adjusted R^2	0.090	0.073	0.101	0.165

Notes: This table contains the results of regressing relative changes in positions in a given ETF by different investor sectors on investor sector dummies. The investment fund sector forms the baseline sector. The unit of observation is on the ETF-investor sector level. The regression specification includes ETF fixed effects that absorb ETF characteristics. In column (2), we restrict attention to equity ETFs. Column (3) contains the results for corporate bond ETFs. Finally, column (4) contains the results for sovereign bond ETFs. Standard errors are clustered at the ETF-level, and the resulting t-statistics are shown in parentheses.
* $p<0.05$.
** $p<0.01$.

We restrict ourselves to the first quarter of 2020 and consider the following regression on the investor type - ETF level to compare the buying and selling of a given ETF share by different investor types:

$$f^{j}_{i,2020Q1} = \sum_{k=1}^{K} \beta_k \times I(Sector_j = k) + \alpha_i + \varepsilon_{i,j}. \tag{2}$$

Here, $I(Sector_j = k)$ is an indicator variable denoting the different investor sectors, where the investment fund sector serves as the baseline. Moreover, we include ETF fixed effects (α_i) to absorb any ETF characteristics. This allows us to compare the differences in the buying and selling behavior across investor sectors within the same ETF during the first quarter of 2020. This identification strategy closely resembles Fricke et al. (2025), who study differential responses to the lagged performance of open-ended equity funds across different investor sectors, and Allaire et al. (2024), who study outflows from open-ended bond funds by different investor sectors during the COVID-19 pandemic.

The results in column (1) of Table 4 are based on a sample of equity, corporate bond, and sovereign bond ETFs. We find that relative to the investment fund sector, net ETF transactions were significantly larger for insurance companies, pension funds, and the residual category of euro area investors. This suggests that investment funds scaled down their ETF holdings to a larger extent than some other key investor sectors, some of which actually increased their holdings in certain types of ETFs (see also Fig. 1(b)). However, we do not find a statistically significant difference between net ETF transactions of investment funds and the household sector. Since the specification for column (1) can mask heterogeneity across ETF types as witnessed in Fig. 1(b), we distinguish between the different ETF types in columns (2) to (4).

Column (2) shows that during the first quarter of 2020, investment funds' net transactions in equity ETFs were 19 percentage points lower compared to households. This effect is also statistically significant. This difference is even more pronounced for some other investor sectors. Specifically, investment funds' net transactions in equity ETFs were 44 percentage lower relative to pension funds, and 79 percentage points lower relative to the other euro area investors. Note that the magnitude of these differences in part reflects net ETF purchases by some investor sectors (see Fig. 1(b)). Column (3) contains the results for corporate bond ETFs. We again find that the reduction in investment funds' ETF positions was on average 20 percentage points larger than the reduction in households' ETF positions, in line with the results for equity ETFs. The difference with respect to insurance companies and pension funds is, however, statistically insignificant. In contrast to the results for equity and corporate bond ETFs in columns (2) and (3), we do not find that investment funds liquidated a larger fraction of their sovereign bond ETF holdings relative to other investor sectors (column (4)).

In summary, we find that for both equity and corporate bond ETFs, investment funds reduced their pre-existing ETF positions relatively more than other investor sectors during the first quarter of 2020. This result is consistent with findings on run behavior in open-ended equity funds by Fricke et al. (2025) and open-ended bond funds by Allaire et al. (2024).

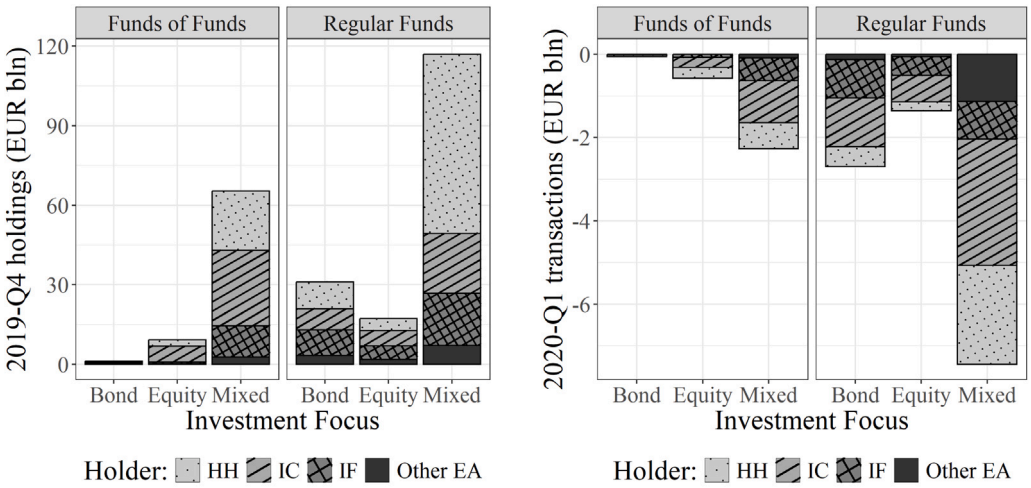
4. Flow-induced ETF sales by open-ended funds

The results in Section 3 suggest that investment funds were the most run-prone investor types in the market for ETFs. One obvious explanation could be that these investment funds sold ETF shares to accommodate redemptions they faced themselves. Table

Table 5
Summary of flows faced by open-ended funds in March 2020.

OEF Type	Fund of Funds	# Funds	% Negative Flow	Distribution Flows (%) 5%	50%	95%
Bond	No	147	80.27	−15.64	−2.29	6.08
Bond	Yes	20	85.00	−6.22	−1.57	1.32
Equity	No	242	66.53	−10.70	−0.77	8.73
Equity	Yes	125	74.40	−13.51	−1.15	2.97
Mixed Assets	No	742	74.66	−9.31	−1.11	3.45
Mixed Assets	Yes	635	69.61	−8.14	−0.94	4.57
Total	–	1911	72.48	−9.91	−1.06	4.52

Notes: This table contains a summary of the flows faced by open-ended funds in March 2020. We distinguish between bond, equity, and mixed assets funds, as well as between regular funds versus funds of funds.



(a) OEF ownership as of 2019-Q4.

(b) Transactions in OEFs during 2020-Q1.

Fig. 5. Ownership and transactions in open-ended funds during 2020-Q1.
Notes: Panel A shows the positions by euro area investors in a set of open-ended funds as of December 2019, broken down by investor type and fund type. The sample of open-ended funds consists of funds belonging to the Undertaking for Collective Investment in Transferable Securities (UCITS) framework that invest in ETFs as of February 2020. Regarding the investor sector, we distinguish between investment funds (IF), households (HH), insurance companies (IC), and the remaining investor types are classified as "Other EA". Panel B shows the net transactions by euro area investors during the first quarter of 2020, broken down by investor type and fund type.
Source: LSEG Lipper, Securities Holdings Statistics.

5 provides a descriptive overview of the flows faced by the open-ended funds in our sample during March 2020. We distinguish between the different types of open-ended funds (bond, equity, and mixed asset funds), as well as between funds classified as "funds of funds" and regular funds that are not. For all subcategories, the percentage of funds in our sample that faced net outflows during March 2020 exceeds 66%. In total, 72% of the open-ended funds in our sample faced net outflows during March 2020. Across subcategories, the median net flow during March 2020 varies between −0.77% and −2.29%, while the 5th and 95th percentiles are indicative of substantial heterogeneity in flows.

Fig. 5 provides insight into the types of investors that were responsible for the outflows faced by the open-ended funds in our sample. Fig. 5(a) shows that the household sector, insurance companies, and investment funds are the main euro area investors in open-ended funds. In particular, households are the largest holder of open-ended bond funds (both funds of funds and regular funds) within the euro area investor base. For open-ended equity funds, insurance companies are the largest euro area holder type (both funds of funds and regular funds). Regarding mixed asset funds, which is the largest type of open-ended fund in our sample, insurance companies are the largest holder sector of those funds that are classified as funds of funds. In contrast, the largest euro area investor type in regular mixed asset funds that are not classified as funds of funds is the household sector. Fig. 5(b) suggests that within the euro area investor base, the largest holder sectors were also the largest net sellers (in absolute terms) of the open-ended investment funds in our sample during the first quarter of 2020. Namely, most of the outflows faced by these open-ended funds stem from insurance companies and the household sector, and to a lesser extent from other investment funds.

Given that the open-ended funds in our sample indeed faced outflows during March 2020, we next test whether open-ended funds that faced larger outflows during March 2020 also sold a larger part of their ETF holdings, relative to open-ended funds that faced smaller outflows. We consider the following regression, as in Ma et al. (2022):

$$Trade_{i,j} = \alpha_j + \beta Flow_i + \gamma' X_i + \varepsilon_{i,j}. \tag{3}$$

Here, the dependent variable is defined as $Trade_{i,j} = \frac{Shares_{i,j}^{March}}{Shares_{i,j}^{Feb}} - 1$, which is the relative change in the number of shares of ETF j held by fund i during March 2020. Fund flows during March 2020 are defined as follows:

$$Flow_{i,t} = \frac{TNA_{i,t} - TNA_{i,t-1} \times (1 + R_{i,t})}{TNA_{i,t-1}}. \tag{4}$$

Here, $TNA_{i,t}$ denotes the fund's total net assets at the end of month t, and $R_{i,t}$ is the fund's return during month t. We also control for a vector of lagged fund characteristics (X_i), which includes the fund's age, the logarithm of TNA, dummy variables indicating whether the fund is a mixed/bond fund (equity funds are the baseline), and a dummy indicating whether the fund is a fund of funds. The inclusion of ETF fixed effects (α_j) implies that we compare differential selling behavior across open-ended funds within the same ETF. By doing so, we implicitly control for ETF characteristics that may be related to funds' selling behavior. This addresses potential endogeneity concerns, as ETF characteristics (such as its geographical investment focus) may be correlated with the magnitude of outflows faced by open-ended funds, as well as their corresponding selling decision. As this identification strategy requires at least two open-ended funds holding a given ETF, we drop ETFs that are held by only one open-ended fund.

Panel A in Table 6 contains the results. In column (1), we consider all types of ETFs in our sample. It follows that within a given ETF, funds that were subject to lower net flows (larger outflows) in March 2020 also sold a significantly larger fraction of their position in this ETF. Specifically, a one percentage point increase in outflows is associated with an additional decrease in ETF holdings by 1.1 percentage points. In columns (2) to (4), we separately consider equity, corporate bond, and sovereign bond ETFs, respectively. Our results continue to hold within each subset of ETFs, although the economic magnitude appears somewhat larger for equity ETFs than for bond ETFs. To rule out the possibility that ETF purchases by open-ended funds that faced inflows confound the results, in Panel B we restrict the sample to open-ended funds that faced net outflows during March 2020. Namely, after facing inflows, funds may not necessarily increase their positions in existing holdings, but can use inflows to open a position in assets that the fund did not hold before. As such, the absolute change in existing positions may be different for outflows versus inflows. In line with this, focusing on the outflow sample leads to stronger results as the coefficient magnitudes have increased. Also, for all types of ETFs, the coefficients are statistically significant. These results are consistent with the hypothesis that redemptions faced by open-ended funds are an important driver of their ETF sales.

The regression specification in Eq. (3) only considers March 2020. To test the external validity of the results in Panels A and B of Table 6, we extend the specification from Eq. (3) over a larger sample period running from January 2019 to June 2024:

$$Trade_{i,j,t} = \alpha_{j,t} + \beta Flow_{i,t} + \gamma' X_{i,t-1} + \varepsilon_{i,j,t}. \tag{5}$$

Since we consider a monthly frequency, the level of observation is now on the fund-ETF-year-month level. The inclusion of ETF × year-month fixed effects ($\alpha_{j,t}$) ensures that we again compare differential selling behavior across open-ended funds within a given ETF within a given month. Panel A of Table 7 contains the results. In line with earlier findings, we find a significantly positive relationship between flows experienced by open-ended funds and their transactions in ETFs. Specifically, the results in column (1) suggest that a one percentage point decrease in net flows (increase in net outflows) is associated with a 1.25 percentage point decrease in ETF holdings. While there is some variation in the coefficient magnitudes across ETF types in columns (2) to (4), all coefficients are positive and statistically significant. We repeat the analysis but now focus on the subset of fund-year-month observations for which net flows are negative, again with the aim to rule out the possibility that the results in Panel A are driven by ETF purchases by funds facing inflows. The results in Panel B confirm our earlier findings as the coefficient estimates are close to the ones in Panel A. As such, the liquidation-to-outflow sensitivity that we find for March 2020 also holds for a more extensive sample period. Nevertheless, the estimated liquidation-to-outflow sensitivity appears larger in March 2020 (Panel B of Table 6) than over the larger sample period (Panel B of Table 7), except for sovereign bond ETFs that in aggregate faced net inflows during March 2020 (see Fig. 3). This may suggest that in March 2020, open-ended funds accommodated a larger fraction of their outflows with ETF sales compared to more general circumstances.

5. Open-ended funds' pecking order of liquidation

Fund managers can employ various liquidity management strategies when accommodating flows. For instance, fund managers can horizontally slice their portfolios in response to outflows, meaning that they sell their most liquid assets first to minimize portfolio rebalancing costs. Alternatively, fund managers can also choose to vertically slice their portfolios, meaning that they sell assets in proportion to their portfolio weights to preserve the fund's portfolio composition. Since both strategies could imply that larger outflows would be associated with larger ETF sales, the results in Section 4 do not directly allow us to make a clear distinction between these different liquidity management strategies. In this section, we therefore test whether in response to outflows, fund managers liquidated a larger fraction of their ETF holdings relative to their holdings in other asset types. If so, this would further suggest that fund managers use ETFs to manage portfolio liquidity. We therefore consider the following regression:

$$Trade_{i,j} = \alpha_i + \beta Flow_i + \gamma' \left(Flow_i \times Type_j\right) + \delta' Type_j + \varepsilon_{i,j}. \tag{6}$$

Table 6
ETF transactions by open-ended funds during March 2020.

Panel A: ETF sales during March 2020				
Dependent variable:	Transaction in ETF j by fund i in March 2020			
ETF Type	All	Equity	Corp. Bond	Sov. Bond
	(1)	(2)	(3)	(4)
Flow	1.14**	1.27**	1.13**	0.907*
	(4.41)	(4.29)	(2.71)	(2.02)
Fund Age	−0.359**	−0.215	−0.691*	−0.598*
	(−2.62)	(−1.38)	(−2.55)	(−2.03)
log(TNA)	−0.293	0.329	−2.26	−2.11*
	(−0.516)	(0.455)	(−1.92)	(−2.09)
Mixed Asset OEF	−6.45**	−7.67**	18.4	12.1
	(−3.07)	(−3.40)	(1.09)	(0.946)
Bond OEF	−5.66	−7.45	22.0	14.1
	(−1.02)	(−0.688)	(1.27)	(0.976)
Fund of Funds	0.467	−1.88	−0.566	7.26
	(0.255)	(−0.862)	(−0.138)	(1.93)
ETF FE	Yes	Yes	Yes	Yes
Observations	8,942	5,678	999	1,276
Adjusted R^2	0.161	0.163	0.145	0.151
Panel B: ETF sales during March 2020 - outflow sample				
Dependent variable:	Transaction in ETF j by fund i in March 2020			
ETF Type	All	Equity	Corp. Bond	Sov. Bond
	(1)	(2)	(3)	(4)
Flow	1.59**	1.65**	1.77**	1.18*
	(6.93)	(5.78)	(3.25)	(2.57)
Fund Age	−0.084	0.054	−0.674*	−0.271
	(−0.636)	(0.357)	(−2.13)	(−0.788)
log(TNA)	−0.822	−0.344	−2.91*	−2.74*
	(−1.20)	(−0.399)	(−1.98)	(−2.05)
Mixed Asset OEF	−6.76**	−8.10**	22.2	8.04
	(−2.76)	(−3.16)	(1.06)	(0.749)
Bond OEF	−5.66	−12.9	30.2	7.52
	(−0.806)	(−0.758)	(1.41)	(0.559)
Fund of Funds	−2.91	−5.57*	−3.85	2.44
	(−1.42)	(−2.23)	(−0.809)	(0.547)
ETF FE	Yes	Yes	Yes	Yes
Observations	6,256	3,968	693	890
Adjusted R^2	0.165	0.172	0.145	0.143

Notes: This table shows the results of regressing monthly changes in open-ended funds' ETF holdings on monthly flows faced by open-ended funds. Specifically, the dependent variable, $Trade_{i,j,t}$, denotes the relative change in the number of shares of ETF j held by fund i during month t. In this table, we focus on March 2020, so the level of observation is on the fund-ETF level. We control for lagged fund characteristics, including fund age, the logarithm of total net assets (TNA), dummy variables indicating mixed asset/bond funds (equity funds provide the baseline), and a dummy variable indicating funds of funds. We also include ETF fixed effects. The t-statistics are based on standard errors clustered at the fund level and are shown in parentheses.
* $p<0.05$.
** $p<0.01$.

Here, the dependent variable is again defined as $Trade_{i,j} = \frac{Shares_{i,j}^{March}}{Shares_{i,j}^{Feb}} - 1$, which is the relative change in the number of shares of security j held by fund i during March 2020. In case security j represents a bond, we consider the relative change in the par value held. This way, our dependent variable is not confounded by valuation changes. The variable $Type_j$ is a categorical variable indicating the type of security j. The inclusion of fund fixed effects (α_i) implies that we compare differential selling behavior across asset types within a given open-ended fund during March 2020. As we focus on funds' pecking order of liquidation in response to outflows, we only consider funds that faced outflows during March 2020.

We first test whether funds disproportionally sold ETF shares relative to other mutual fund shares, which represent the largest portfolio component of the open-ended funds in our sample (see Fig. 2). Note that some funds may avoid mutual fund shares altogether and invest in ETFs instead. In such a scenario, where mutual fund holdings are substituted by holdings in ETFs, it is not possible to test whether a fund disproportionally sold ETF shares relative to mutual fund shares. We therefore restrict attention to funds holding both mutual fund shares and ETF shares. The resulting subsample includes 1167 open-ended funds that hold both mutual fund shares and ETFs. Panel A of Table 8 contains the results of estimating equation (6) in which attention is restricted to observations where $Type_j$ corresponds to mutual fund shares and ETFs, where mutual fund shares serve as the baseline. Funds may have many positions with tiny portfolio weights. Consequently, small absolute changes in these portfolio holdings may lead to extreme relative changes. We therefore consider a weighted least squares approach in columns (2) and (5), where the weight of

Table 7
ETF transactions by open-ended funds during 2019–2024.

Panel A: monthly ETF sales during 2019–2024				
Dependent variable:	Transaction in ETF j by fund i in month t			
ETF Type	All (1)	Equity (2)	Corp. Bond (3)	Sov. Bond (4)
Flow	1.25**	1.24**	1.42**	1.35**
	(35.5)	(27.6)	(24.3)	(24.4)
Fund Age	0.0003	0.007	0.037	−0.017
	(0.022)	(0.467)	(1.43)	(−0.738)
log(TNA)	−0.212**	−0.326**	−0.310**	0.069
	(−3.48)	(−4.49)	(−2.62)	(0.669)
Mixed Asset OEF	−0.132	−0.283	5.64**	2.19*
	(−0.461)	(−0.950)	(3.28)	(2.03)
Bond OEF	−0.355	−0.516	5.36**	2.10
	(−0.683)	(−0.705)	(2.97)	(1.64)
Fund of Funds	−0.462*	−0.455*	−1.02*	0.283
	(−2.32)	(−1.97)	(−2.53)	(0.823)
ETF × Year-Month FE	Yes	Yes	Yes	Yes
Observations	707,087	453,615	86,140	93,869
Adjusted R^2	0.161	0.165	0.128	0.156
Panel B: monthly ETF sales during 2019–2024 - outflow sample				
Dependent variable:	Transaction in ETF j by fund i in month t			
ETF Type	All (1)	Equity (2)	Corp. Bond (3)	Sov. Bond (4)
Flow	1.26**	1.27**	1.49**	1.41**
	(27.7)	(22.3)	(15.6)	(16.0)
Fund Age	−0.001	0.003	0.047	−0.017
	(−0.088)	(0.212)	(1.44)	(−0.576)
log(TNA)	−0.281**	−0.425**	−0.331	−0.044
	(−3.65)	(−4.80)	(−1.92)	(−0.308)
Mixed Asset OEF	−0.010	−0.201	6.31*	2.86*
	(−0.030)	(−0.563)	(2.46)	(2.01)
Bond OEF	−0.507	0.377	5.65*	2.25
	(−0.738)	(0.542)	(2.10)	(1.35)
Fund of Funds	−0.861**	−0.892**	−1.72**	0.314
	(−3.73)	(−3.38)	(−3.24)	(0.708)
ETF × Year-Month FE	Yes	Yes	Yes	Yes
Observations	385,702	240,421	50,424	52,871
Adjusted R^2	0.167	0.175	0.134	0.159

Notes: This table shows the results of regressing monthly changes in open-ended funds' ETF holdings on monthly flows faced by open-ended funds. Specifically, the dependent variable, $Trade_{i,j,t}$, denotes the relative change in the number of shares of ETF j held by fund i during month t. We consider a sample period covering 2019–2024, so the level of observation is on the fund-ETF-year-month level. We control for lagged fund characteristics, including fund age, the logarithm of total net assets (TNA), dummy variables indicating mixed asset/bond funds (equity funds provide the baseline), and a dummy variable indicating funds of funds. We also include ETF × year-month fixed effects. The t-statistics are based on standard errors clustered at the fund level and are shown in parentheses.

* $p<0.05$.

** $p<0.01$.

each observation is determined by the portfolio weight that fund i held in security j at the end of February 2020. In columns (3) and (6), we focus on a subsample where we restrict the sample to observations where the pre-existing portfolio weight exceeds 0.5%.

It follows from columns (1) to (3) in Panel A of Table 8, in which fund fixed effects are excluded, that open-ended funds also pass on their outflows to other mutual funds as the coefficient on standalone flows is significantly positive. This result is in line with recent findings by Fricke and Wilke (2023), Allaire et al. (2024), and Fricke et al. (2025). Moreover, it also follows that on average, ETF sales were significantly larger than sales of mutual fund shares, as indicated by the standalone coefficient on $Type_j = ETF$. For instance, the result in column (3) shows that funds scaled down their ETF holdings by 5.09 additional percentage points relative to other mutual fund shares, all else equal. Moreover, the coefficients on the interaction between fund flows and $Type_j = ETF$ suggest that in response to an increase in outflows during March 2020, funds scaled down their ETF holdings by more than their holdings in mutual fund shares. Only when using the weighted least squares approach does the coefficient on the interaction term in column (2) become statistically insignificant. Columns (4) to (6) contain the results when fund fixed effects are included, implying that we compare the liquidation of different assets within the same open-ended fund. The results from this more restrictive specification confirm the results from columns (1) to (3). Overall, these results provide some evidence that funds relied relatively more on ETF shares than on other mutual fund shares when accommodating outflows during March 2020.

Table 8
Selling of ETFs relative to other portfolio assets during March 2020.

Panel A. Mutual fund shares versus ETFs						
Dependent variable:	Transaction in asset j by fund i in March 2020					
Sample:	Full	Full	$w > 0.5\%$	Full	Full	$w > 0.5\%$
Model:	OLS	WLS	OLS	OLS	WLS	OLS
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	−7.52**	−7.38**	−8.17**			
	(−8.74)	(−10.7)	(−9.61)			
Flow	1.01**	1.17**	1.18**			
	(6.38)	(7.89)	(6.53)			
$Type_j = ETF$	−5.23**	−6.44**	−5.09**	−3.73**	−3.47**	−3.37**
	(−3.99)	(−4.58)	(−3.92)	(−3.17)	(−2.99)	(−2.92)
$Flow \times Type_j = ETF$	0.725**	0.388	0.543*	0.841**	0.442	0.657*
	(2.85)	(1.37)	(2.13)	(3.13)	(1.78)	(2.53)
Fund FE	No	No	No	Yes	Yes	Yes
Observations	20,843	20,843	18,051	20,843	20,843	18,051
Adjusted R^2	0.020	0.032	0.024	0.185	0.234	0.205
Panel B. Equity versus Equity ETFs						
Constant	−2.91**	−4.18**	−4.10**			
	(−3.16)	(−4.59)	(−3.95)			
Flow	1.26**	1.17**	1.25**			
	(6.28)	(7.80)	(6.46)			
$Type_j = Eq.\ ETF$	−3.49	−1.65	−3.42	1.63	1.31	−0.690
	(−1.56)	(−0.738)	(−1.54)	(0.764)	(0.602)	(−0.308)
$Flow \times Type_j = Eq.\ ETF$	0.689	0.667	0.675	1.25*	1.43**	1.37**
	(1.36)	(1.34)	(1.21)	(2.55)	(2.65)	(2.62)
Fund FE	No	No	No	Yes	Yes	Yes
Observations	63,202	63,202	11,253	63,202	63,202	11,253
Adjusted R^2	0.018	0.025	0.025	0.145	0.209	0.199
Panel C. Corporate bonds versus Corporate Bond ETFs						
Constant	−10.9*	−3.08*	−2.41			
	(−2.14)	(−2.56)	(−1.97)			
Flow	0.752	1.03**	0.897**			
	(1.41)	(6.28)	(5.03)			
$Type_j = CB\ ETF$	0.944	−8.04*	−6.49	−10.4*	−9.61*	−7.33
	(0.154)	(−2.45)	(−1.51)	(−2.49)	(−2.35)	(−1.49)
$Flow \times Type_j = CB\ ETF$	2.01*	1.22	1.50	1.44	1.43	1.59
	(2.17)	(1.36)	(1.27)	(1.70)	(1.59)	(1.34)
Fund FE	No	No	No	Yes	Yes	Yes
Observations	19,511	19,511	2,631	19,511	19,511	2,631
Adjusted R^2	0.008	0.041	0.033	0.219	0.159	0.115
Panel D. Sovereign bonds versus Sovereign Bond ETFs						
Constant	−12.0**	−10.2**	−12.2**			
	(−3.21)	(−4.78)	(−5.82)			
Flow	0.839*	1.39**	1.23**			
	(2.42)	(5.33)	(4.43)			
$Type_j = SB\ ETF$	−5.09	−6.20	−5.78	−7.58	−8.74	−7.92
	(−0.912)	(−1.07)	(−1.03)	(−1.39)	(−1.30)	(−1.35)
$Flow \times Type_j = SB\ ETF$	1.76*	0.444	1.39	1.96	0.744	1.39
	(2.26)	(0.625)	(1.63)	(1.97)	(0.842)	(1.46)
Fund FE	No	No	No	Yes	Yes	Yes
Observations	4,757	4,757	1,800	4,757	4,757	1,800
Adjusted R^2	0.016	0.034	0.032	0.146	0.200	0.138

Notes: This table presents the results of regressing changes in open-ended funds' portfolio holdings in March 2020 on fund flows faced in March 2020, interacted with a dummy indicating the asset class of the corresponding holding. The dependent variable denotes the %-change in the number of shares of asset j held by fund i during March 2020. Panel A focuses on funds that hold both ETFs as well as mutual fund shares, Panel B on funds that invest in both equities as well as equity ETFs, and Panel C on funds that invest in both corporate bonds as well as corporate bond ETFs. Standard errors are clustered at the fund level and the corresponding t-statistics are shown in parentheses.

* $p<0.05$.

** $p<0.01$.

We next examine the sales of equity ETFs relative to the sales of direct equity holdings. We therefore focus on funds holding both equities and equity ETF shares. The resulting subsample includes 511 open-ended funds. Panel B of Table 8 contains the results of estimating equation (6) in which attention is restricted to observations where $Type_j$ corresponds to equities and equity ETFs, where equities serve as the baseline. It follows from columns (1) to (3), in which fund fixed effects are excluded, that open-ended funds also pass on their outflows to the equity market as the coefficient on standalone flows is significantly positive, which is in accordance with Coval and Stafford (2007) and Lou (2012). Moreover, we do not detect a significant difference between the sales of equity ETFs versus the sales of equities based on the standalone coefficient on $Type_j$. Also, the coefficients on the interaction between fund flows and $Type_j$ are statistically insignificant, so based on this specification we fail to reject the hypothesis that funds disproportionally scaled down their equity ETF holdings relative to direct equity investments during March 2020. However, the results from columns (4) to (6), in which we include fund fixed effects, suggest that funds scaled down their equity ETF holdings by more than their direct equity holdings in response to an increase in outflows during March 2020. As such, the results in columns (4) to (6) suggest that funds relied relatively more on equity ETF shares than direct equity holdings when accommodating outflows during March 2020.

In Panel C of Table 8, we compare the sales of corporate bond ETFs versus the sales of direct corporate bond holdings. Some corporate bonds matured in March 2020, and hence changes in funds' holdings of these bonds equal −100%. We therefore exclude observations corresponding to bonds that matured in March 2020 as these do not reflect active selling decisions. Given the relative illiquidity of corporate bonds, corporate bond ETFs may be a particularly appealing instrument to manage fund flows. Nevertheless, our sample contains relatively few funds (only 150) that simultaneously invest in both corporate bonds and corporate bond ETFs. The results in Panel C do not provide robust evidence that funds relied more on corporate bond ETFs than on direct corporate bond holdings when meeting outflows as the interaction coefficients are statistically insignificant. Still, the standalone coefficient on the corporate bond ETF dummy is negative in columns (2) to (6) and statistically significant in columns (2), (4), and (5). This suggests that open-ended funds liquidated a larger share of their ETF holdings than their direct corporate bond holdings, irrespective of the level of outflows.

We next compare the sales of sovereign bond ETFs versus the sales of direct sovereign bond holdings in Panel D of Table 8. Our sample consists of 179 open-ended funds that simultaneously hold sovereign bonds and sovereign bond ETFs. Again, the standalone coefficient on the monthly flow in March 2020 is significantly positive in columns (1) to (3), suggesting that open-ended funds sold both sovereign bonds and sovereign bond ETFs in response to flows. Although the coefficients on the interaction between a fund's net flow during March 2020 and the sovereign bond ETF dummy are all positive, we only find statistical significance in column (1).

Overall, after controlling for fund fixed effects, we find the strongest results for equity ETFs. Within a given open-ended fund, the liquidation-to-outflow sensitivity is significantly larger for equity ETFs than for direct equity holdings.

We finally test the external validity of our results over a longer sample period by extending the specification in Eq. (6) as follows:

$$Trade_{i,j,t} = \alpha_{i,t} + \beta Flow_{i,t} + \gamma' \left(Flow_{i,t} \times Type_j \right) + \delta' Type_j + \varepsilon_{i,j,t}. \tag{7}$$

Here, the unit of observation is on the fund-asset-year-month level, where the sample spans the period between January 2019 and June 2024. Table 9 contains the results, where we again focus on fund-month combinations in which the net monthly flow is negative. While the coefficient magnitudes have decreased somewhat compared to the results for March 2020 in Table 8, all coefficients on the interaction between fund flows and the ETF dummies are now statistically significant. This suggests that during a broader sample period, investment funds disproportionally scale down their ETF holdings in response to outflows relative to other types of holdings. As noted before, we find our strongest results for equity ETFs when focusing on March 2020. During the broader sample period, the magnitudes of the interaction coefficients are somewhat smaller. This suggests that the economic magnitude of the difference in the liquidation-to-outflow sensitivity of equity ETFs vis-à-vis direct equity holdings was larger in March 2020 than during the broader sample period.

Concluding, we find that funds passed on their outflows to various asset types, and this effect tends to be more pronounced for holdings in ETFs. While reaching conclusive evidence on the exact ranking of ETFs in funds' pecking order of liquidation is challenging, our results imply that large aggregate outflows from open-ended funds can be an important driver behind secondary market ETF sales by investment funds.

6. Impact of investment fund ownership on ETF markets

In this section, we study primary market flows in the ETF market in relation to the ownership composition of ETFs. Primary market flows refer to the creation and redemption of ETF shares.[4] Only so-called authorized participants (APs) are able to create and redeem ETF shares, as other investors in ETFs (including investment funds) can only trade ETF shares in secondary markets. As such, sales of ETFs by market participants other than APs do not directly translate in primary market flows for the corresponding ETFs. However, if some big investors and/or many small investors simultaneously sell ETF shares, this leads to downwards pressure on the price of the ETF in secondary markets. This, in turn, can lead to a situation where the ETF's secondary market price drops below the ETF's net asset value (NAV), which creates an arbitrage opportunity for APs. If APs subsequently redeem ETF shares to pocket the difference between the ETF's price and the NAV, ETF sales by investors in secondary markets ultimately translate into primary market outflows for the ETF in question.

[4] We refer to Lettau and Madhavan (2018) for a more detailed explanation of the ETF creation-redemption process.

Table 9
Selling of ETFs relative to other portfolio assets - sample period 2019–2024.

Panel A. Mutual fund shares versus ETFs						
Dependent variable:	Transaction in asset j by fund i in month t					
Sample:	Full	Full	$w > 0.5\%$	Full	Full	$w > 0.5\%$
Model:	OLS	WLS	OLS	OLS	WLS	OLS
	(1)	(2)	(3)	(4)	(5)	(6)
Flow	0.950**	1.14**	1.00**			
	(22.5)	(33.8)	(28.1)			
$Type_j = ETF$	−1.48**	−1.83**	−1.59**	−1.63**	−1.39**	−1.60**
	(−9.61)	(−10.6)	(−10.9)	(−12.0)	(−11.2)	(−12.2)
$Flow \times Type_j = ETF$	0.280**	0.190**	0.237**	0.217**	0.149**	0.197**
	(5.19)	(3.64)	(5.05)	(4.53)	(3.21)	(4.17)
Year-Month FE	Yes	Yes	Yes	No	No	No
Fund × Year-Month FE	No	No	No	Yes	Yes	Yes
Observations	1,189,427	1,189,427	1,026,606	1,189,427	1,189,427	1,026,606
Adjusted R^2	0.010	0.018	0.012	0.086	0.132	0.087
Panel B. Equity versus Equity ETFs						
Flow	0.865**	1.01**	1.05**			
	(9.43)	(16.3)	(14.6)			
$Type_j = Eq.\ ETF$	−1.21**	−1.66**	−1.94**	−0.391	−1.19**	−1.81**
	(−3.79)	(−5.52)	(−6.31)	(−1.35)	(−3.96)	(−6.01)
$Flow \times Type_j = Eq.\ ETF$	0.508**	0.357**	0.319**	0.402**	0.500**	0.417**
	(4.11)	(2.89)	(2.65)	(3.64)	(4.19)	(3.66)
Year-Month FE	Yes	Yes	Yes	No	No	No
Fund × Year-Month FE	No	No	No	Yes	Yes	Yes
Observations	3,254,233	3,254,233	624,007	3,254,233	3,254,233	624,007
Adjusted R^2	0.006	0.014	0.014	0.108	0.164	0.143
Panel C. Corporate bonds versus Corporate Bond ETFs						
Flow	0.312	0.864**	0.805**			
	(0.976)	(9.65)	(9.80)			
$Type_j = CB\ ETF$	2.70**	0.803*	0.698	0.322	0.216	0.405
	(2.94)	(2.09)	(1.66)	(0.780)	(0.522)	(1.01)
$Flow \times Type_j = CB\ ETF$	1.40**	0.833**	0.878**	1.09**	1.07**	1.07**
	(4.07)	(4.43)	(5.61)	(5.80)	(5.46)	(5.88)
Year-Month FE	Yes	Yes	Yes	No	No	No
Fund × Year-Month FE	No	No	No	Yes	Yes	Yes
Observations	1,026,243	1,026,243	153,180	1,026,243	1,026,243	153,180
Adjusted R^2	0.026	0.017	0.014	0.343	0.172	0.126
Panel D. Sovereign bonds versus Sovereign Bond ETFs						
Flow	0.596**	1.07**	1.04**			
	(3.75)	(11.4)	(9.98)			
$Type_j = SB\ ETF$	1.77*	2.43**	2.03**	2.28**	2.16**	2.13**
	(2.30)	(4.61)	(4.18)	(5.13)	(3.75)	(4.51)
$Flow \times Type_j = SB\ ETF$	1.08**	0.776**	0.683**	0.864**	0.704**	0.636**
	(5.53)	(5.04)	(4.24)	(4.92)	(4.05)	(3.93)
Year-Month FE	Yes	Yes	Yes	No	No	No
Fund × Year-Month FE	No	No	No	Yes	Yes	Yes
Observations	249,140	249,140	99,374	249,140	249,140	99,374
Adjusted R^2	0.017	0.021	0.018	0.211	0.186	0.113

Notes: This table presents the results of regressing monthly changes in open-ended funds' portfolio holdings on contemporaneous monthly fund flows, interacted with a dummy indicating the asset class of the corresponding holding. The sample period covers 2019–2024. The dependent variable denotes the %-change in the number of shares of asset j held by fund i during March 2020. Panel A focuses on funds that hold both ETFs as well as mutual fund shares, Panel B on funds that invest in both equities as well as equity ETFs, and Panel C on funds that invest in both corporate bonds as well as corporate bond ETFs. Standard errors are clustered at the fund level and the corresponding t-statistics are shown in parentheses.

* $p<0.05$.

** $p<0.01$.

The results in previous sections have shown that outflows from open-ended funds were passed on to ETFs during March 2020. This may have a material impact on primary market ETF flows through increased arbitrage opportunities for APs, given the large ownership stake of investment funds in the ETF market. We therefore hypothesize that ETFs with higher investment fund ownership faced larger primary market outflows relative to ETFs with lower investment fund ownership. To reduce the impact of potential measurement error and noise in daily flows (Choi et al., 2022), we aggregate daily primary market net flows to a weekly frequency

as follows Badidi et al. (2025):

$$f_{i,d+1} = \ln\left(\frac{TNA_{i,d+1}}{TNA_{i,d} \times (1 + R_{i,d+1})}\right) \tag{8}$$

$$f_{i,d+1:d+5} = \ln\left(\frac{TNA_{i,d+5}}{TNA_{i,d} \times (1 + R_{i,d+1:d+5})}\right) = \sum_{s=1}^{5} f_{i,d+s}. \tag{9}$$

Here, $f_{i,d+1}$ denotes the logarithm of the daily net primary market flow for ETF i, and $f_{i,d+1:d+5}$ denotes the logarithm of the cumulative weekly net primary market flow. Using this definition, the weekly net primary market flow is identical to the sum of the corresponding daily net primary market flows. We multiply the resulting value by 100 to express primary market flows in percentages for ease of interpretation.

To assess the impact of higher ex ante investment fund ownership on ETFs' primary market flows during stress periods, we consider the following panel regression:

$$f_{i,q,t} = \alpha_t + \delta_i + \gamma' \left(Crisis_t \times IFOwnership_{i,q-1}\right) + \delta IFOwnership_{i,q-1} + \lambda' Controls_{i,t-1} + \varepsilon_{i,q,t}. \tag{10}$$

The dependent variable $f_{i,q,t}$ denotes the weekly primary market flow of ETF i during week t in quarter q as defined in Eq. (9). The variable $Crisis_t$ is a vector of five weekly dummy variables corresponding to the five weeks spanning the COVID-19 crisis phase (February 24–March 27). Moreover, $IFOwnership_{i,q-1}$ denotes the fraction of ETF i that is held by euro area investment funds per the end of the previous quarter $(q-1)$. As mentioned in Section 2, we restrict the sample to ETFs owned for at least 75% by euro area investors. This filter ensures that we observe the sectoral composition for a sufficient fraction of the ETF's TNA, as SHS coverage is low for holdings by investors domiciled outside the euro area.[5] α_t and δ_i denote time and ETF fixed effects, respectively. The vector of controls includes the lagged weekly return, the lagged weekly primary market flow, the logarithm of lagged total net assets, and the ETF's age. Controlling for lagged fund performance is important, as it is a key driver of fund flows (see, e.g., Berk and Green, 2004). While the positive relation between lagged performance and flows has typically been attributed to investors using lagged performance to infer fund manager skill, Dannhauser and Pontiff (2025) find a significantly positive flow-performance relationship for index-tracking equity ETFs as well, for which fund manager skill is less relevant. These results suggest that return chasing by investors is not limited to active mutual funds. Controlling for lagged performance is therefore important when examining primary market flows. In doing so, we aim to alleviate the concern that primary market outflows from ETFs are driven by changing expectations about future returns, to the extent these can be explained by lagged ETF performance.

Panel A of Table 10 contains the results based on the regression specification in Eq. (10). We are mainly interested in the interactions between the crisis dummies and lagged investment fund ownership. For equity ETFs (column (1)), primary market flows for ETFs with high investment fund ownership responded stronger immediately after the shock relative to ETFs with low investment fund ownership, as suggested by the significantly negative coefficient corresponding to the first crisis week. In contrast, primary market flows during the week between February 28 and March 6 are significantly positively related to investment fund ownership. A possible explanation for this result may be that institutional investors, including investment funds, responded faster to the initial COVID-19 outbreak than other types of investors (hence the negative coefficient corresponding to the first crisis week). If other investors responded with a lag (relative to investment funds), this could explain why relative primary market flows of ETFs with high investment fund ownership were larger during the week between February 28 and March 6. In addition, the impact of the COVID-19 crisis on primary market flows appears to be more persistent for equity ETFs with relatively high investment fund ownership given the significantly negative coefficients on the interactions with the third, fourth, and fifth crisis week dummies. The economic magnitude is largest for the weekly primary market flow between March 13 and March 20. During this week, an increase in investment fund ownership by one standard deviation is associated with an increase in weekly primary market outflows of 1.18 percentage points ($28.32\% \times -4.17$).

For corporate bond ETFs (column (2)), we find that primary market outflows for ETFs with high investment fund ownership were persistently larger during all five crisis weeks relative to ETFs with low investment fund ownership, with the effect being statistically significant during the second, fourth, and fifth crisis weeks. We again find that the economic magnitude is largest during the week between March 13 and March 20. During this week, an increase in investment fund ownership by one standard deviation is associated with a 0.88 percentage point increase in weekly primary market outflows ($25.49\% \times -3.44$).

Finally, column (3) contains the results for sovereign bond ETFs. Higher investment fund ownership was initially associated with higher primary market flows in the first two weeks of the COVID-19 crisis, although this effect is only statistically significant for the second crisis week. In contrast, weekly primary market flows for ETFs with high investment fund ownership were significantly lower in the weeks between March 6–March 13 and March 13–March 20. Specifically, during the third crisis week (March 6–March 13), primary market outflows for ETFs with high investment fund ownership were on average 1.86 percentage points higher ($25.68\% \times -7.23$). This result differs from our results for equity and corporate bond ETFs. A potential explanation is that, during the first two weeks of the crisis, open-ended funds with large allocations to sovereign bonds and sovereign bond ETFs may have faced lower outflows or maybe even inflows due to investors flying to safety. Such lower redemption requests may subsequently have reduced the need to sell sovereign bond ETFs. This could potentially explain the higher primary market flows (or lower primary market outflows) for sovereign bond ETFs with relatively high investment fund ownership in the first two crisis weeks.

[5] Our results are robust to using a 50% threshold.

Table 10
Investment fund ownership and ETF primary market flows.

Dependent variable:	Primary market flow of ETF i during week t		
ETF Type:	Equity	Corp. Bond	Sov. Bond
	(1)	(2)	(3)
Panel A. Investment fund ownership as a % of TNA			
%IF × Week 1 [Feb 21–Feb 28]	−0.824**	−0.709	0.159
	(−4.40)	(−1.44)	(0.187)
%IF × Week 2 [Feb 28–Mar 6]	1.17**	−1.12*	2.29**
	(4.44)	(−2.40)	(4.44)
%IF × Week 3 [Mar 6–Mar 13]	−2.58**	−0.495	−7.23**
	(−9.74)	(−1.07)	(−7.87)
%IF × Week 4 [Mar 13–Mar 20]	−4.17**	−3.44**	−5.24**
	(−22.1)	(−6.77)	(−5.65)
%IF × Week 5 [Mar 20–Mar 27]	−2.12**	−3.25**	0.841
	(−8.96)	(−6.11)	(1.75)
ETF FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	29,226	4,047	8,285
Adjusted R^2	0.046	0.060	0.061
Panel B. Above-median investment fund ownership dummy			
IF dummy × Week 1 [Feb 21–Feb 28]	−1.34**	−0.961**	0.571
	(−13.0)	(−3.72)	(1.76)
IF dummy × Week 2 [Feb 28–Mar 6]	0.134	−1.60**	1.68**
	(1.17)	(−6.41)	(7.93)
IF dummy × Week 3 [Mar 6–Mar 13]	−1.43**	−0.339	−2.61**
	(−11.9)	(−1.35)	(−7.62)
IF dummy × Week 4 [Mar 13–Mar 20]	−2.43**	−3.75**	−1.59**
	(−23.4)	(−14.5)	(−4.67)
IF dummy × Week 5 [Mar 20–Mar 27]	−0.714**	−2.60**	0.527**
	(−6.17)	(−9.64)	(2.76)
ETF FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	29,226	4,047	8,285
Adjusted R^2	0.047	0.062	0.061

Notes: This table shows the results of regressing ETFs' weekly primary market flows on lagged investment fund ownership interacted with crisis dummies. In Panel A, we consider investment fund ownership as a percentage of the ETF's net asset value ($\%IF$). In Panel B, we consider a dummy variable that equals 1 if investment fund ownership is above the median investment fund ownership for ETFs of the same type. The unit of observation is on the ETF-week level and the sample period covers January 2019 up to and including June 2020. The sample consists of equity ETFs in column (1), corporate bond ETFs in column (2), and sovereign bond ETFs in column (3). We control for lagged weekly returns, lagged weekly primary market flows, the logarithm of total net assets, and ETF age. All specifications include time and ETF fixed effects. Standard errors are clustered at the week level, and t-statistics are shown in parentheses.

* $p<0.05$.

** $p<0.01$.

Panel B of Table 10 contains the results of an alternative specification where we replace the continuous investment fund ownership variable by a dummy variable that equals 1 for ETFs with above-median investment ownership in the corresponding quarter. Here, we compute median investment fund ownership for equity ETFs, corporate bond ETFs, and sovereign bond ETFs separately for each quarter. This alternative specification confirms our earlier findings. On average, primary market outflows for equity ETFs with above-median investment fund ownership were 2.43 percentage points higher relative to equity ETFs with below-median investment fund ownership during the fourth crisis week (March 13–March 20). For corporate bond ETFs, this difference is even larger at 3.75 percentage points. Regarding sovereign bond ETFs, we find that primary market flows for ETFs with above-median investment fund ownership were 2.61 percentage points lower relative to ETFs with below-median investment fund ownership during the third crisis week (March 6–March 13). However, in line with the results in Panel A, we find that during the second crisis week (February 28–March 6) and fifth crisis week (March 20–March 27), primary market flows were actually higher for ETFs with above-median investment fund ownership.

Overall, our results are consistent with Cai et al. (2025) and Dannhauser and Pontiff (2025), who both argue that institutional investors are key drivers of ETF primary market flows. The ultimate impact of ETF selling pressure in secondary markets on the ETF's underlying assets depends on a number of factors. First, ETF sales in secondary markets do not need to translate one-to-one into ETF share redemptions, as other market participants may absorb (part of) the ETF sales by investment funds. Second, ETF sponsors have discretion when setting the composition of creation/redemption baskets, which are not always a pro-rata slice of the

ETF's holdings. Especially for ETFs investing in less liquid markets, such as the corporate bond market, ETF sponsors may employ "fractional baskets" that only contain a subset of the ETF's holdings (Shim and Todorov, 2023; Koont et al., 2025). As such, the composition of redemption baskets moderates the impact of ETF sales by investment funds on underlying markets. Third, after the AP has redeemed an ETF share and received the assets from the redemption basket, the AP faces the decision of whether or not to sell the assets in secondary markets. In case the AP sells the underlying assets in secondary markets, the sale of the ETF ultimately translates into a sale of the assets in the redemption basket. However, in case the AP decides to keep the assets on its balance sheet, the AP may effectively prevent a fire sale of the underlying assets (Shim and Todorov, 2023). Studying the impact of ETF sales on underlying markets therefore requires information on the composition of redemption baskets, as well as granular high-frequency information on the AP's balance sheets, which is unavailable to us.

7. Conclusion

In this paper, we investigate how open-ended investment funds use ETFs to manage liquidity during periods of stress and how their trading affects ETF markets. We find that investment funds are the largest group of ETF investors in the euro area. Our results from fixed-effects panel regressions in which we compare the behavior from different investor types invested in the same ETF, within the same quarter, show that investment funds were selling more ETF shares than any other investor type during the COVID-19 pandemic. We then show that ETF selling by open-ended funds during March 2020 was stronger for funds facing larger outflows and that funds relied relatively more on selling of ETF shares than other assets to accommodate redemption requests. These results not only hold during March 2020, but also for a longer sample period covering 2019–2024. This is consistent with funds using ETFs for managing liquidity and raising cash, if needed, during stress and normal times. Finally, we also find that a larger share of fund ownership in ETFs translates into more sizable redemptions in the primary market for equity and corporate bond ETFs during the COVID-19 crisis.

Selling ETF shares in stressed markets allowed open-ended funds to preserve larger cash positions and holdings of other assets. At the same time, they were passing on the increased liquidity demand to the ETF secondary and primary markets, and ultimately to the liquidity providers in underlying securities markets. The results are consistent with open-ended funds passing on part of their outflows to the ETFs that they are invested in. Our findings provide insight into a possible contagion channel from traditional open-ended funds to ETF markets. They also highlight the importance of robust and well-functioning ETF markets for the stability of the broader mutual fund sector, especially during periods of stress.

The impact of passing on these liquidity demands to ETFs for underlying securities markets is still an open question. This depends on the extent to which the sales of ETF shares in secondary markets translate into ETF share redemptions in the primary market, on the composition of redemption baskets as set by the ETF sponsor, as well as on the behavior by APs. Unfortunately, we do not have access to the data required to study this question, and we believe this provides a promising avenue for future research.

CRediT authorship contribution statement

Lennart Dekker: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Luis Molestina Vivar:** Writing – review & editing, Methodology, Conceptualization. **Christian Weistroffer:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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